INORGANIC CUMULATIVE EXAM
April 14, 2000

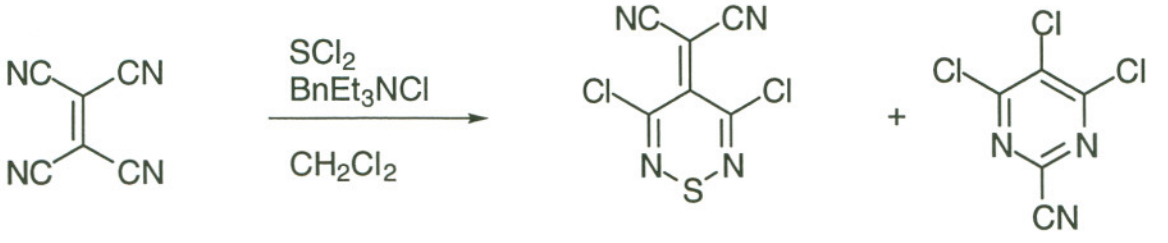
Electronic spin is an important consideration in the chemistry of transition metals. Rationalize, in detail, the following observations in terms of the electronic properties of the complexes.

1. In solution, many complexes of Fe(II) that contain simple ligands such as Cl^-, water and ammonia have high magnetic moments. Other complexes such as those containing phenanthroline ligands are diamagnetic. Combinations of these ligands produce complexes that have strongly temperature dependent magnetic moments. The moments can be near zero at low temperatures and rise quickly at high temperatures.

2. The intensities of the absorption bands of many transition metals are very weak with extinction coefficients on the order of 1-100. Simple aqueous solutions of manganese(II) show bands that are particularly weak with extinction coefficients very much less than 1.

3. Most of the transition metals complexes have excited states with lifetimes on the pico or femtosecond time scale. A few complexes have absorption bands with very large extinction coefficients and excited-state lifetimes on the order of nanoseconds.

Draw reasonable reaction mechanisms for the formation of both products.



PART B

Each of the questions below deals with some aspect of the chemistry of metal carbonyl compounds.

1) Give two different methods for the synthesis of "binary" metal carbonyl compounds. (Only CO ligands)

2) Suggest a synthesis beginning with a readily available starting material (i.e. commercially available from Aldrich, Strem, etc.) for each of the following.

   a) $HMn(CO)_5$

   b) a neutral, tetrahedral complex with CO and NO ligands

   c) $Fe_3(CO)_{12}$

   d) $(CH_3CO)Mn(CO)_5$

3) Account for the preference for metal-to-carbon bonding in terminal MCO linkages.

4) In addition to the linear terminal bonding mode, what other types of M-CO bonding are found in metal carbonyl complexes?